

Lecture 30

4/6/26
JRA

Contrast in Projection Radiography

Beer's Law

$$I(x) = I_0 e^{-\mu x}$$

μ = linear attenuation coefficient (cm^{-1})

Bigger μ = faster attenuation

Distances
measured
in cm
in med.
imaging

$$I_0 \rightarrow \begin{array}{|c|c|c|} \hline x_1 & x_2 & x_3 \\ \hline \mu_1 & \mu_2 & \mu_3 \\ \hline \end{array} \rightarrow I(x)$$

What comes out

Goal: start wrapping up Radiography
next = nuclear imaging

Projection radiography is in contrast to CT

Last week did activity on contrast
Today do derivation

Basic idea is having attenuating (absorbing)
material in x-ray beam

Intensity decays exponentially → GO TO NOTES
AT TOP OF
PAGE

$$I(x) = I_0 e^{-\mu_1 x_1} e^{-\mu_2 x_2} \dots e^{-\mu_n x_n}$$

$$= I_0 e^{-(\mu_1 x_1 + \mu_2 x_2 + \dots + \mu_n x_n)}$$

Local Contrast of an Absorbing Object in a Background

Punch Line $C_{\text{local}} = 1 - e$

Start w/ ^{suitable} defn of contrast

Assume object is more attenuating than bkgd

$$C_{\text{local}} = \frac{I_B - I_0}{I_0}$$

$$C_{\text{local}} = \begin{cases} 0 & I_0 = I_B \\ 1 & I_0 = 0 \end{cases}$$

object completely absorbing



$$C_{\text{local}} = \frac{I_B - I_0}{I_B}$$

Want to know intensity on exit

$$C_{\text{local}} = \frac{\cancel{I_0} e^{-\mu_B x_B} - \cancel{I_0} e^{-(\mu_0 x_0 - \mu_B (x_B - x_0))}}{\cancel{I_0} e^{-\mu_B x_B}}$$

$$= e^{-(\mu_0 x_0) - \mu_B x_0}$$

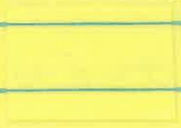
$$= e^{-(\mu_0 - \mu_B) x_0}$$

Key Implications

Local contrast close to 1 if $e^{-(\mu_o - \mu_b)x_o}$

$$1 - e^{-(\mu_o - \mu_b)x_o}$$

x_o big $\Rightarrow$ contrast better

(1)  Big x_o ^{attenuating object} improves contrast Doctor inherits this value

(2) Big $\mu_o - \mu_b$ meaning μ_o & μ_b very different

Easier to see object if it attenuates differently than bkg.

Further implications

(a) μ of bone & ST are very different, especially at low energy

Doctor can manipulate energy of beam
 μ_s are energy dependent

- More diff at low E
- More similar at high E

Bone shows up well against ST, especially at low E

(b) Contrast degrades as energy of beam increases because μ s become more similar

(c) Turns out that doctor can enhance contrast w/ contrast agents

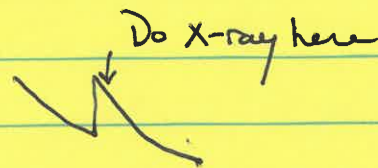
These are excreted by kidneys

Iodine - used for blood	} high Z elements
Barium - used for GI tract	

Iodine is in blood improves soft tissue contrast

This is because absorption $\propto Z^3$

K shell absorption edge



Exploiting absorption edge subject of last activity

How about resolution? NOT diffraction limited

Pipe Dream

$$\lambda_{\text{X-Ray}} \approx 1 \text{ nm}$$

so 1 not a problem
 ~~$\lambda/2$~~ NO

Reality

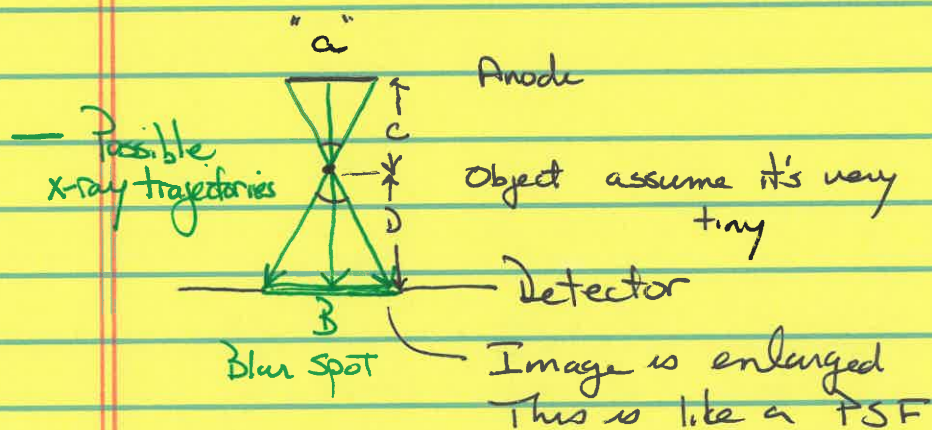
Real resolution is determined by other parameters
Can be as low as $\approx 0.1 \text{ nm}$

What limits resolution?

Finite size of X-ray source (anode)
Symbol of source size "a"

Anode creates X-rays

Electrons hit metal
Some energy creates
X-rays, some
creates heat.
Can't be too tiny



C = distance between
anode & object

D = distance between
object & detector

These are similar isosceles triangles

$$\frac{a}{c} = \frac{B}{D}$$

$$B = \frac{aD}{c}$$

How do you improve resolution?

$$B = \frac{aD}{C}$$

Good res small a, D
big C

Small "a" improves resolution

Another easy thing to control:

small "D" improves resolution

Big "C" good too

Want small "a" but heating problems limit the smallness

Workaround rotate anode so using different parts

Small $D \Rightarrow$ use contact radiography whenever possible (e.g., put hand on detector)
 $D \rightarrow \emptyset$

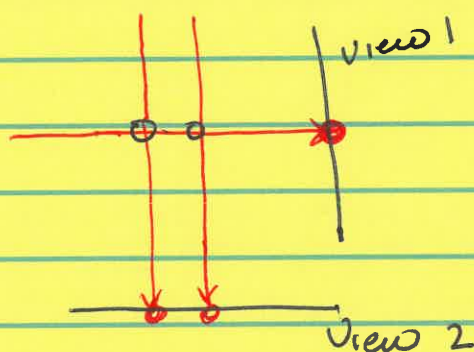
Lots of scattering

Use grids to reduce

Computed Tomography

Projections, or views, give lots of information but they suffer from superposition

Limitation of projections / views

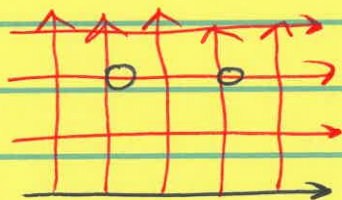


Can't distinguish
 ○ ○
 ∞
 ○

Different views have ~~different~~ ^{complementary} information

Early CT scanner

Source of pencil beam = very thin beam



Translate source & detector that moves in synchrony
 Rotate to get another view in 1° increments
 Hounsfield went thru 180° (180 views)

You can reconstruct ^{an actual 2D slice of} the object w/ all these views

Need to repeat for additional slices

Computer-based algorithm

This was called translate-rotate scanner (because both)

Modern scanners are rotate-rotate
Fan beam

Sequential $\rightarrow$ Helical scans

Algorithms tested on phantom